A map is an equivalence as soon as each point of its codomain supplies a proof that it is one

The Univalent Foundations Program

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Editorial note. The counted statement is the author cor environment at source lines 481-483 and its proof environment opens on line 484, the next line; the pairing is by adjacency and the retained proof range 484-487 is the whole environment.

Editorial note. The book label-number dump gives this label as 4.4.6 (env cor, chapter 4, section 4, sixth counter value), so the counters chapter 4, section 4 and thm 5 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and proof refer to no label outside the card: the fibre criterion and the homotopy fibre are standard material of this chapter, recorded as prerequisites rather than treated as delegations.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: a map whose codomain pointwise supplies an equivalence proof is an equivalence (author label thm:equiv-inhabcod, book Corollary 4.4.6).

Allowed standard prerequisites (editorial).

- Equivalence through contractible fibres. A map is an equivalence exactly when each of its homotopy fibres is contractible.. It converts the goal into the pointwise contractibility of the homotopy fibres, which is how the proof proceeds.
- Homotopy fibres of a map. The homotopy fibre of a map at a point of the codomain is the type of points of the domain together with an identification of their image with that point.. It is the type whose contractibility is established for an arbitrary point of the codomain.

Corollary 4.4.6. *If $f : A \to B$ is such that $B \to \mathsf{isequiv}(f)$, then f is an equivalence.*

Proof. To show f is an equivalence, it suffices to show that $\mathsf{fib}_f(y)$ is contractible for any $y : B$. But if $e : B \to \mathsf{isequiv}(f)$, then given any such y we have $e(y) : \mathsf{isequiv}(f)$, so that f is an equivalence and hence $\mathsf{fib}_f(y)$ is contractible, as desired. □